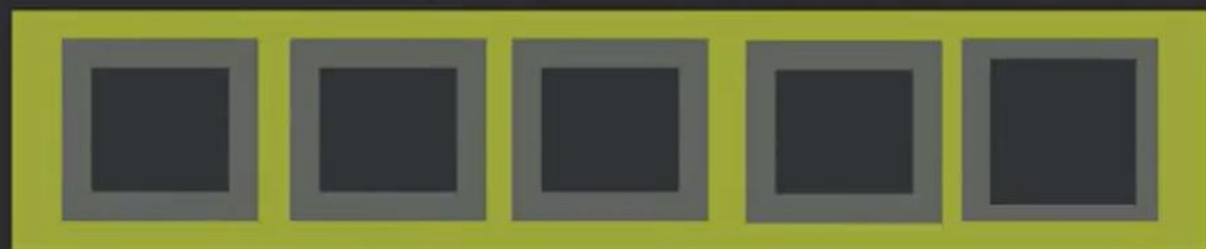
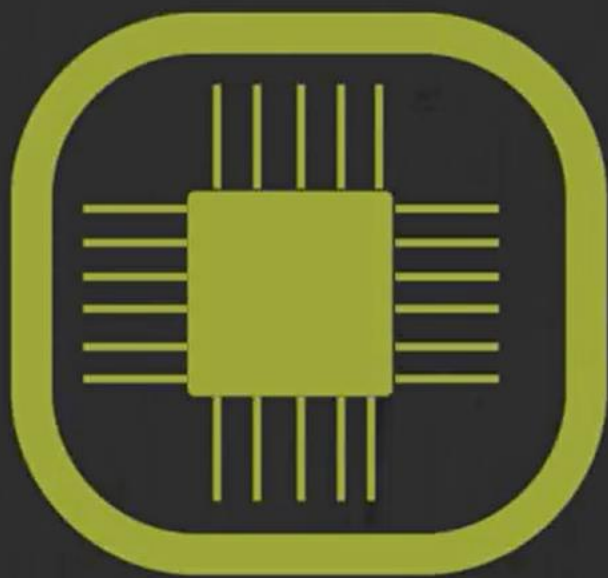


SRAM vs DRAM



RAM (Random Access Memory)

```
graph TD; RAM[RAM (Random Access Memory)] --- SRAM[SRAM (Static RAM)]; RAM --- DRAM[DRAM (Dynamic RAM)];
```

SRAM (Static RAM)

DRAM (Dynamic RAM)

SRAM (Static RAM)

DRAM (Dynamic RAM)

Usage

Cache Memory

Main Memory

Speed

Very Fast

Fast

Cost

Costly

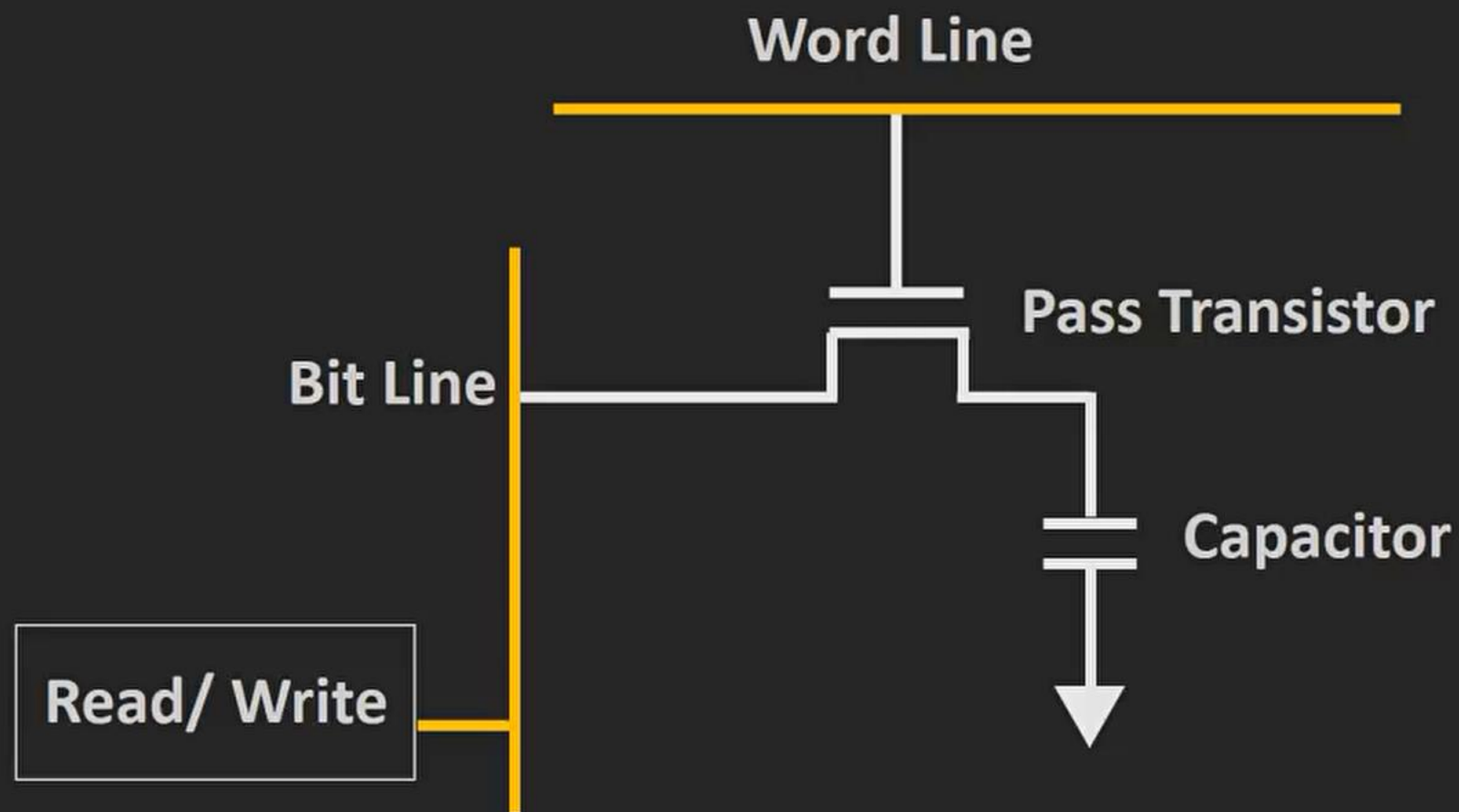
Cheaper than SRAM

Density

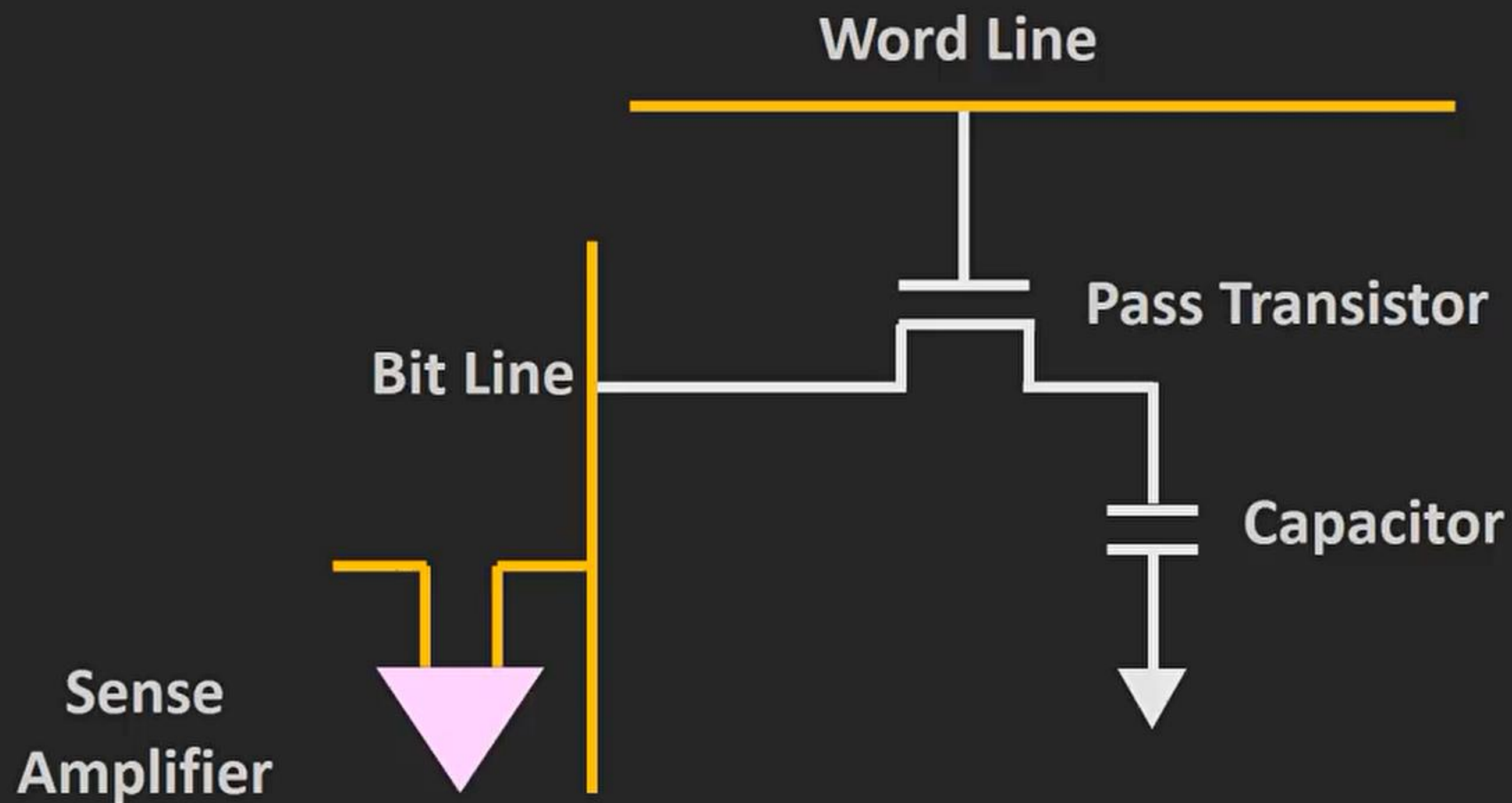
Low

High

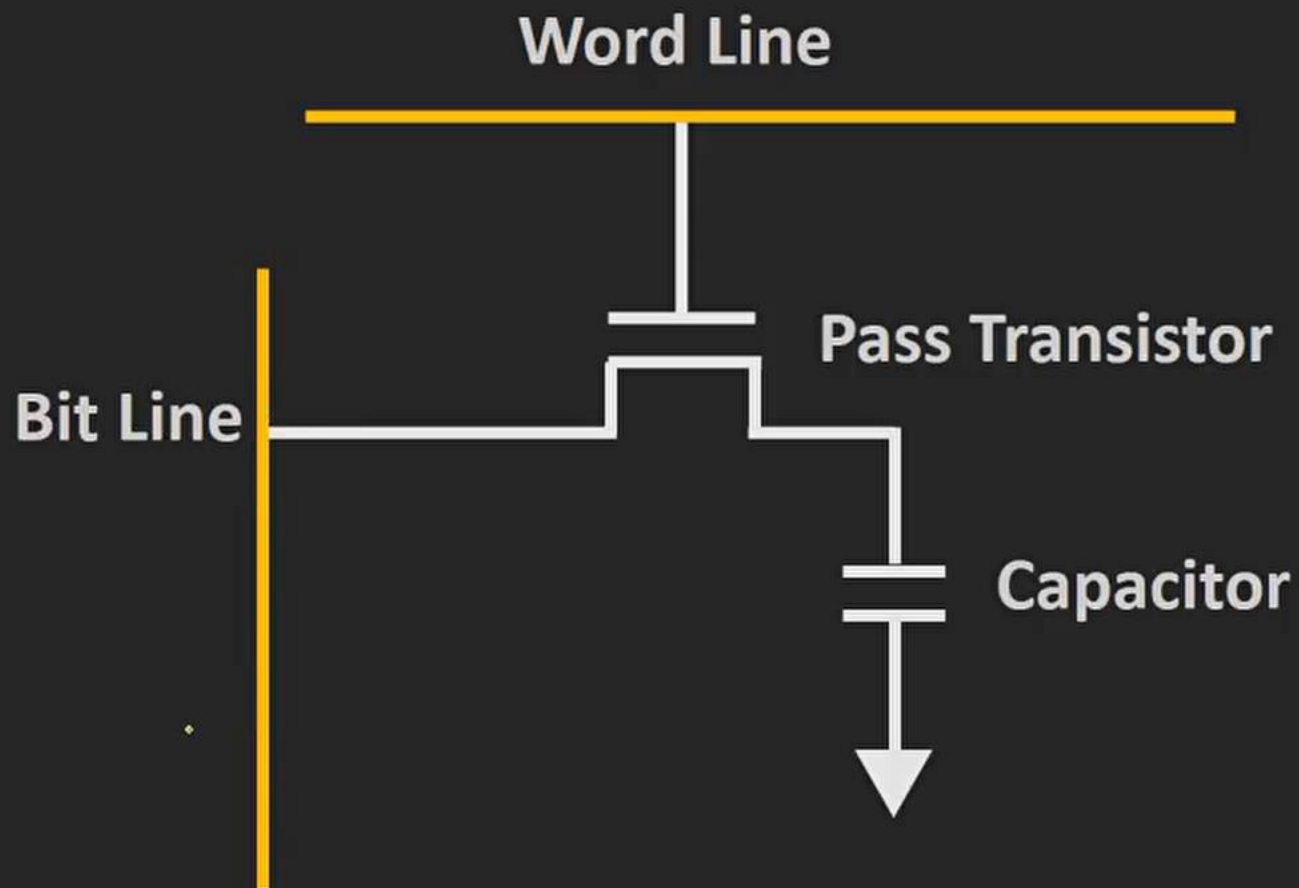
DRAM Cell



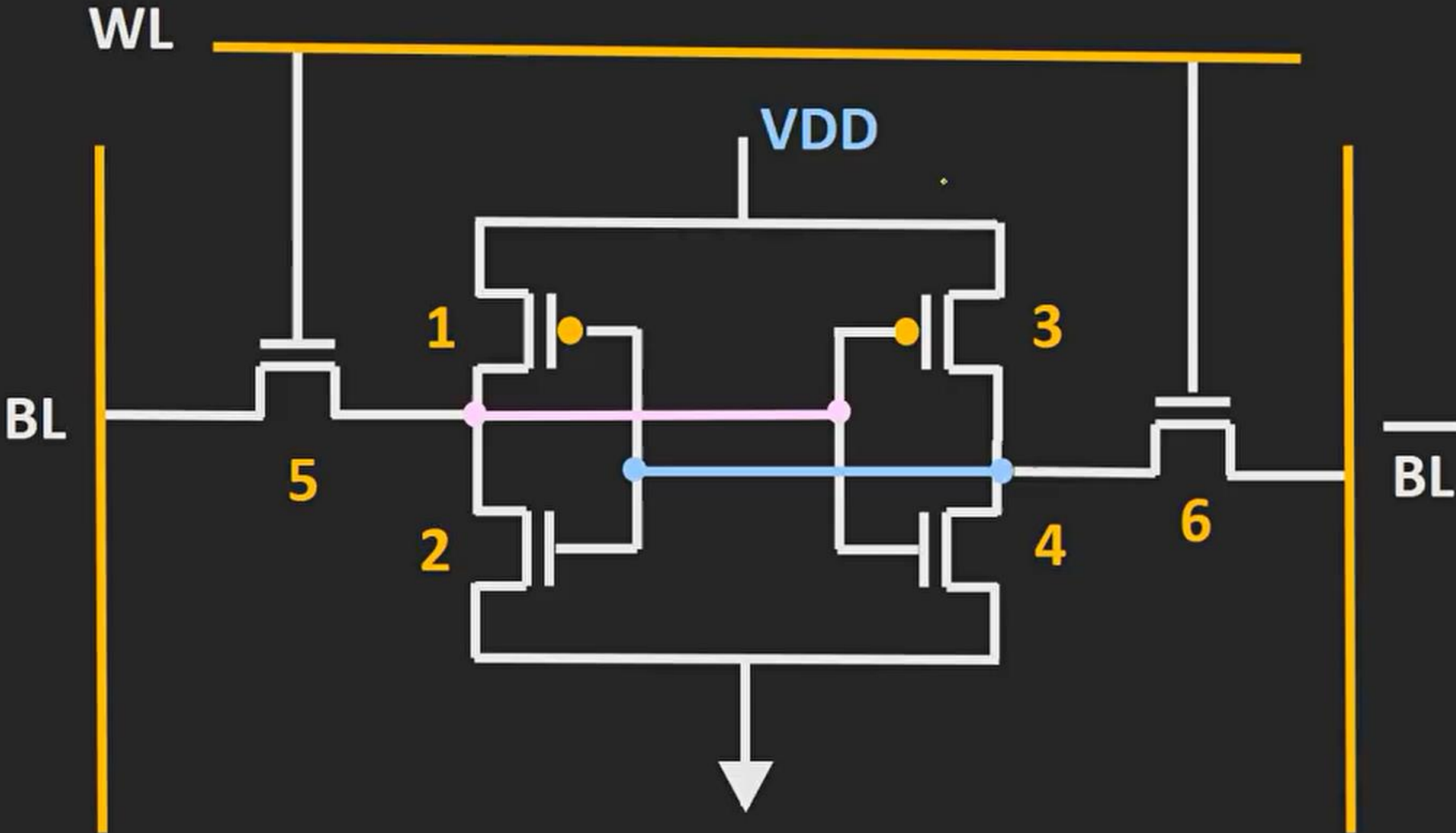
DRAM Read Operation



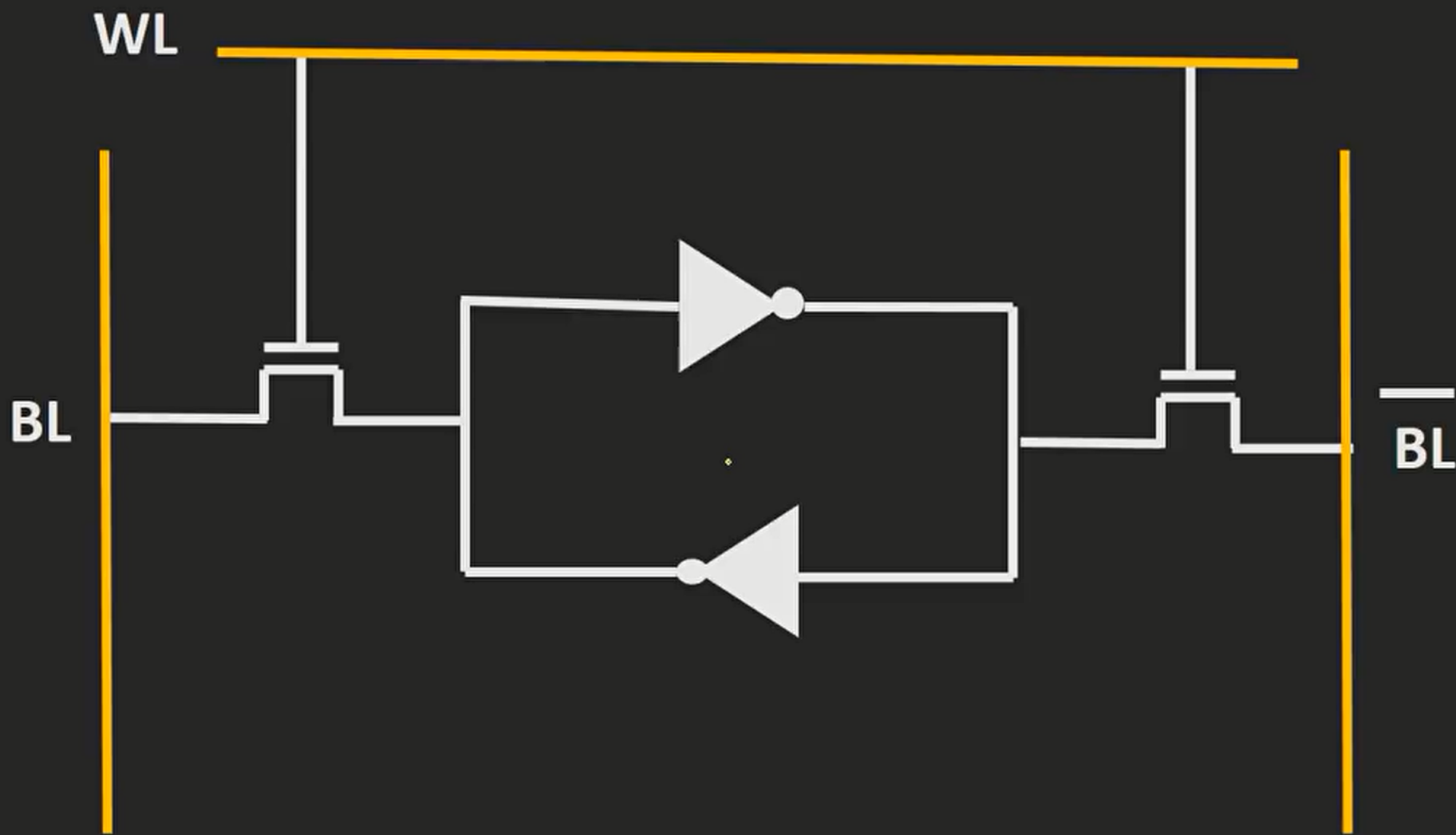
DRAM Write Operation



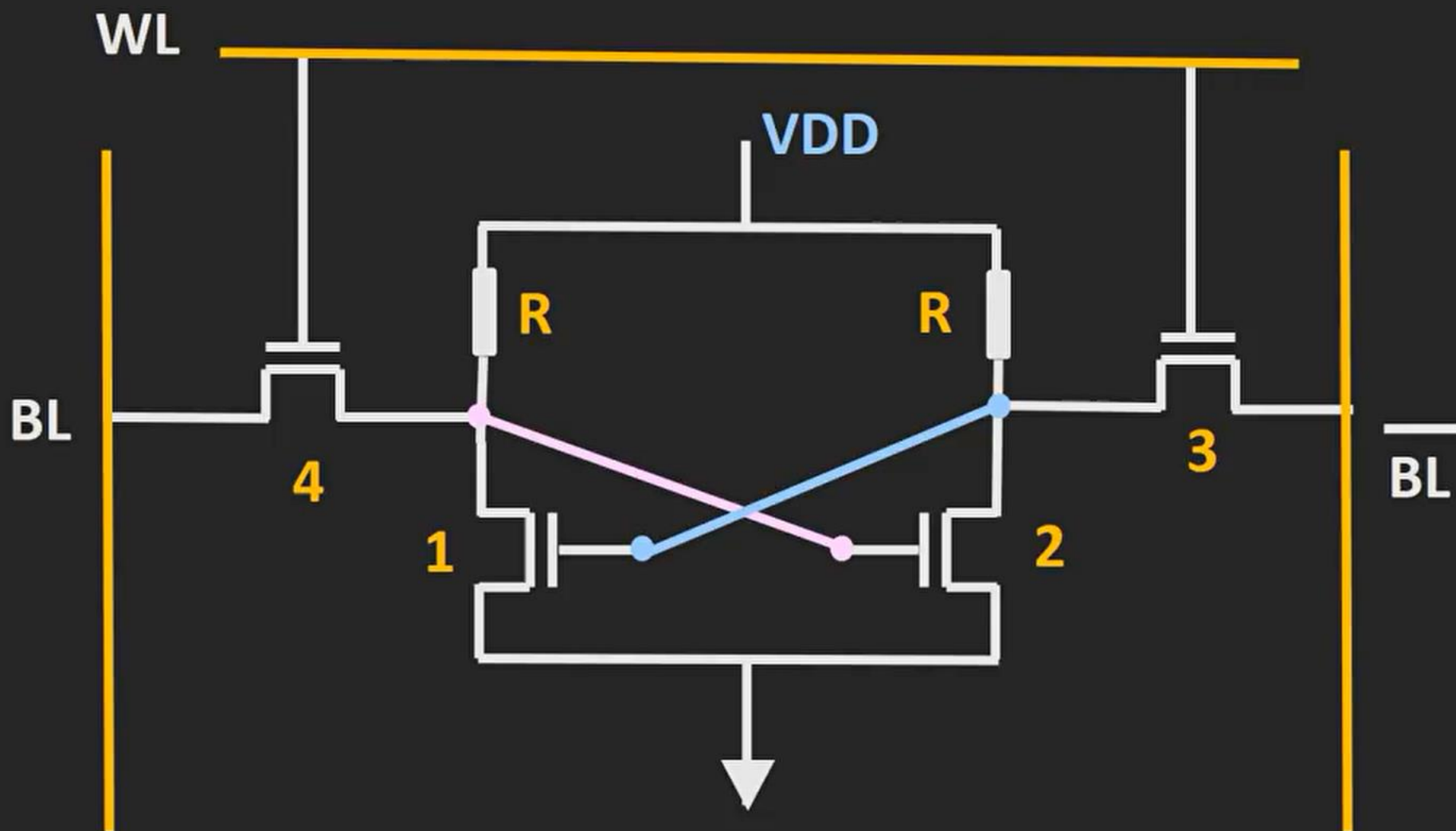
SRAM Cell (6T)



SRAM Read Operation



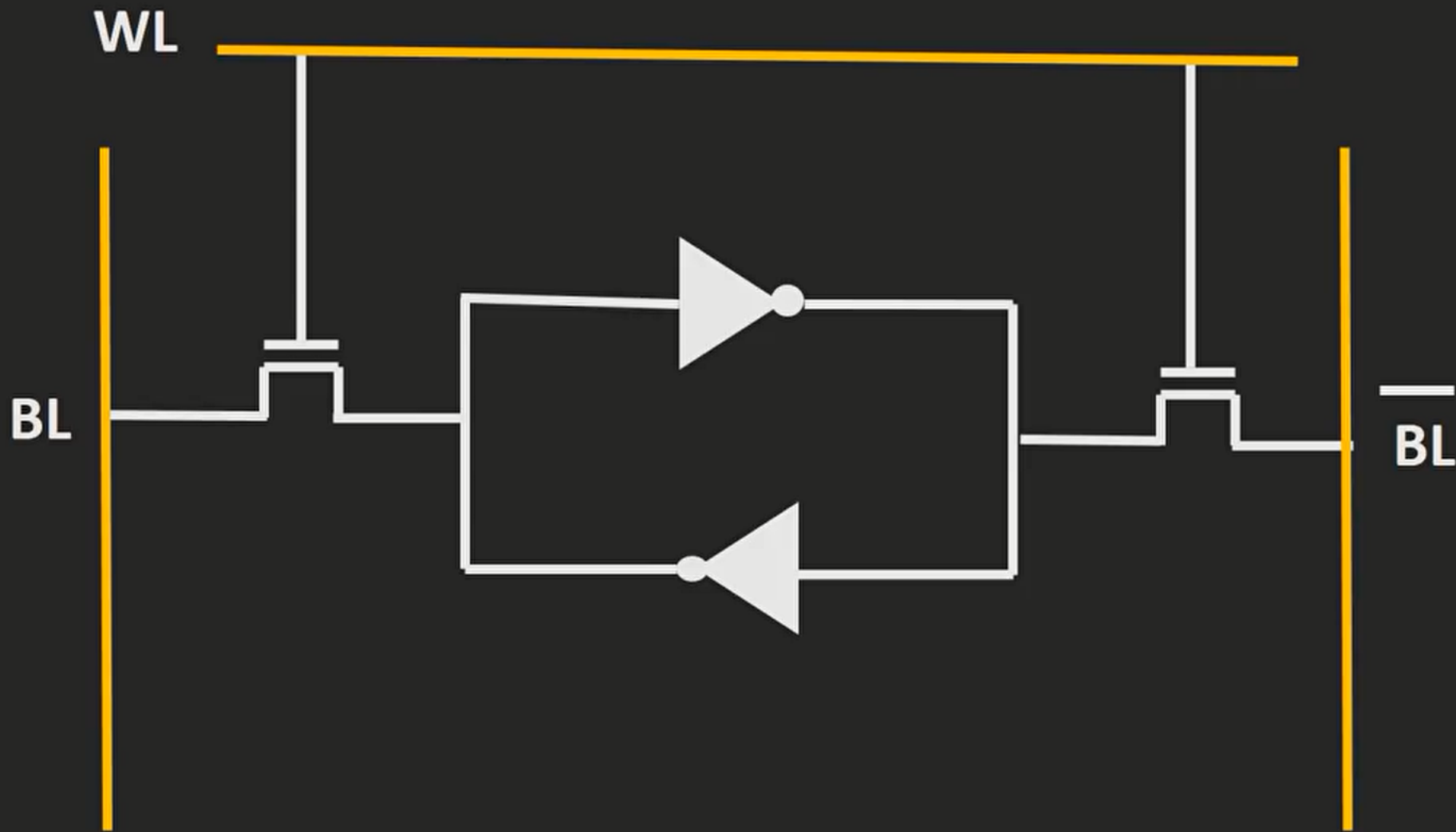
SRAM Cell (4T)



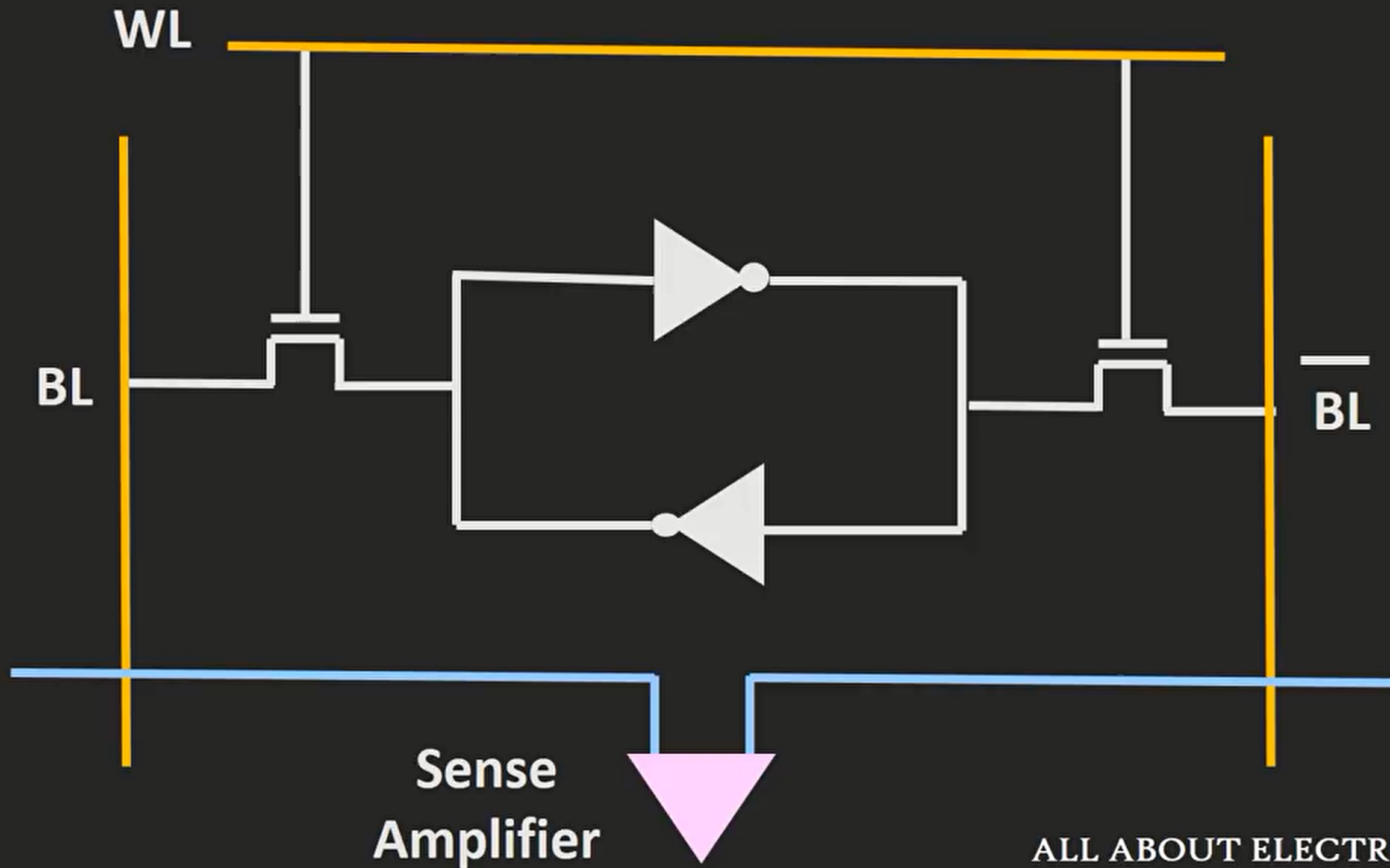
SRAM – 6X space than DRAM

Cost per bit Increases

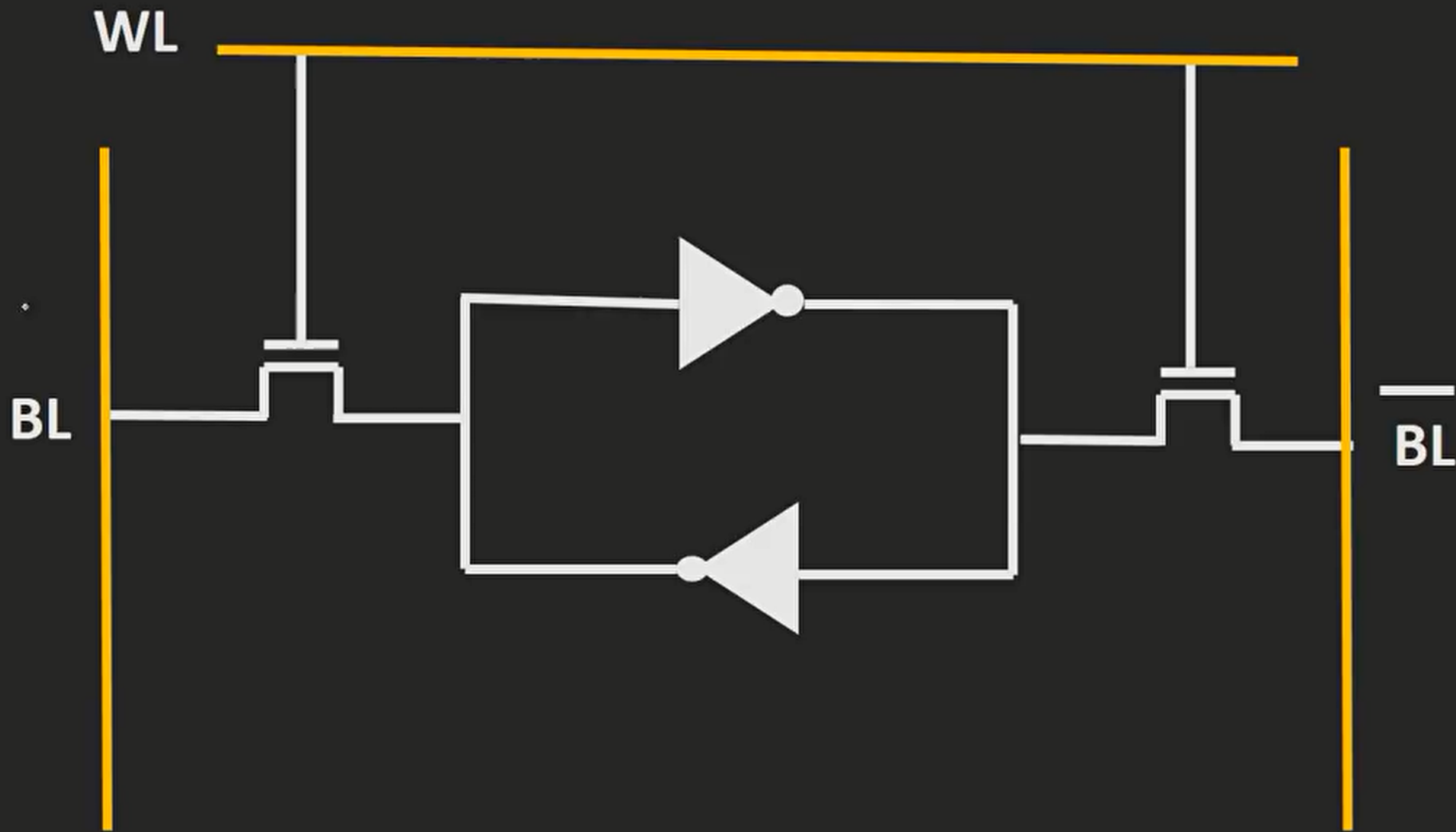
SRAM Read Operation



SRAM Read Operation



SRAM Write Operation



SRAM (Static RAM)

DRAM (Dynamic RAM)

Usage

Cache Memory

Main Memory

Speed

Very Fast

Fast

Cost

Costly

Cheaper than SRAM

Density

Low

High